

method. The first eight strings in the `waterDisplay` list fill the eight-liter bucket, the next five strings the five-liter bucket, and the final three strings the three-liter bucket.

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1. """Water Bucket Puzzle, by Al Sweigart al@inventwithpython.com
2. A water pouring puzzle.
3. More info: https://en.wikipedia.org/wiki/Water\_pouring\_puzzle
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, game, math, puzzle"""
6.
7. import sys
8.
9.
10. print('Water Bucket Puzzle, by Al Sweigart al@inventwithpython.com')
11.
12. GOAL = 4 # The exact amount of water to have in a bucket to win.
13. steps = 0 # Keep track of how many steps the player made to solve this.
14.
15. # The amount of water in each bucket:
16. waterInBucket = {'8': 0, '5': 0, '3': 0}
17.
18. while True: # Main game loop.
19.     # Display the current state of the buckets:
20.     print()
21.     print('Try to get ' + str(GOAL) + 'L of water into one of these')
22.     print('buckets:')
23.
24.     waterDisplay = [] # Contains strings for water or empty space.
25.
26.     # Get the strings for the 8L bucket:
27.     for i in range(1, 9):
28.         if waterInBucket['8'] < i:
29.             waterDisplay.append(' ') # Add empty space.
30.         else:
31.             waterDisplay.append('WWWWWW') # Add water.
32.
33.     # Get the strings for the 5L bucket:
34.     for i in range(1, 6):
35.         if waterInBucket['5'] < i:
36.             waterDisplay.append(' ') # Add empty space.
37.         else:
38.             waterDisplay.append('WWWWWW') # Add water.
39.
40.     # Get the strings for the 3L bucket:
41.     for i in range(1, 4):
42.         if waterInBucket['3'] < i:
43.             waterDisplay.append(' ') # Add empty space.
44.         else:
45.             waterDisplay.append('WWWWWW') # Add water.
46.
47.     # Display the buckets with the amount of water in each one:
48.     print('')
49.     8|{7}|
50.     7|{6}|
```